Do Now · retrieval

Lesson 120 · Following Algorithms and Building Trace Tables

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What drives decisions in spiral development?

Q2

Why can an array provide direct indexed access?

Q3

A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

Q4

For the binary value 1010, what is its decimal value?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify state — Find variables, data structures, conditions and inputs.

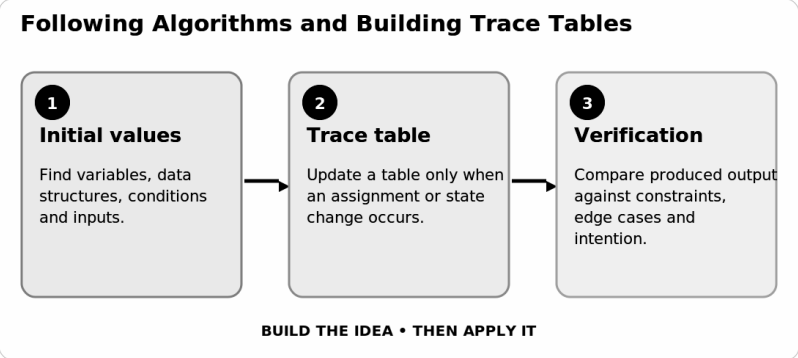
Core idea 2

Trace each step — Update a table when an assignment or state change occurs. For a recursive function, record each call , its argument and its eventual return value; unlike a loop, calls wait on the call stack.

Core idea 3

Check outcome — Compare output against constraints and edge cases. Recursion must reach a base case that returns without another recursive call, or it cannot terminate safely.

Key visual · complete concept



New Learning · applying the ideas

Vocabulary · use the correct technical terms

trace table — Record of algorithm state as steps execute.

iteration — Repeated execution of instructions.

dry run — Manual simulation of algorithm.

boundary test — Test at edge of valid input domain.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace total ← 0; FOR n = 1 TO 4: IF n MOD 2 = 0 THEN total ← total + n; NEXT n; OUTPUT total.

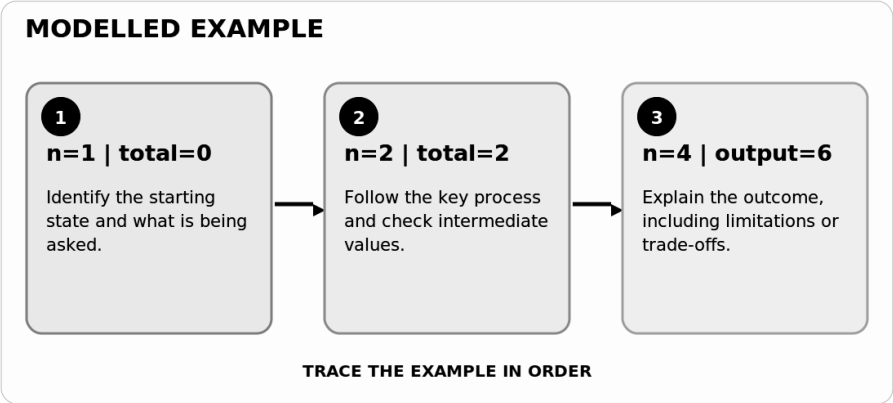
Follow the worked method

1. Model answer — At n=1 total 0; n=2 total 2; n=3 total 2; n=4 total 6; output 6. Record n and total at each loop step. — Check: link each claim to the scenario and use precise OCR terminology.

2. Trace calls before returns — SumTo(3) waits for SumTo(2) , then SumTo(1) , then SumTo(0) . The first three calls remain active until the base case returns.

3. Follow the returns in reverse order — SumTo(0)=0 ; then SumTo(1)=1 , SumTo(2)=3 , SumTo(3)=6 . An iterative alternative would use a loop and running total; recursion instead retains pending calls.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace count ← 0; FOR i = 1 TO 5: IF i MOD 2=1 THEN count←count+1; NEXT; OUTPUT count. [6]

Question 2

Explain why initialising max ← 0 can cause a fault when finding the maximum of negative values. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Trace recursive SumTo(4) using the method shown, name its base case and explain one difference from an iterative solution. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A trace table only needs the final value of each variable.

Use key vocabulary from memory

1. Explain trace table in your own words.

2. Explain iteration in your own words.

3. Explain dry run in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.